

Soumia KHAY
MBDCC
2025/2026

TP 4: Développement d'un Serveur TCP Multi-Thread avec Pool de Threads

```
import java.io.*;
import java.net.*;
import java.text.SimpleDateFormat;
import java.util.Date;
import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;

public class MultiThreadServer {
    public static void main(String[] args) {
        int port = 5000;
        ExecutorService pool = Executors.newFixedThreadPool(5);

        try (ServerSocket serverSocket = new ServerSocket(port)) {
            System.out.println("Serveur démarré sur le port " + port);

            while (true) {
                Socket clientSocket = serverSocket.accept();
                System.out.println("Client connecté : " + clientSocket.getInetAddress().getHostAddress());
                pool.submit(new ClientHandler(clientSocket));
            }
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

Soumia KHAY
MBDCC
2025/2026

// Classe interne pour gérer les clients

```
class ClientHandler implements Runnable {
```

```
    private Socket clientSocket;
```

```
    public ClientHandler(Socket socket) {
```

```
        this.clientSocket = socket;
```

```
    }
```

```
@Override
```

```
public void run() {
```

```
    String threadName = Thread.currentThread().getName();
```

```
    System.out.println("Thread " + threadName + " traite le client " +
```

```
        clientSocket.getInetAddress().getHostAddress());
```

```
    try (
```

```
        BufferedReader in = new BufferedReader(new  
        InputStreamReader(clientSocket.getInputStream()));
```

```
        PrintWriter out = new PrintWriter(clientSocket.getOutputStream(), true)
```

```
    ) {
```

```
        String message;
```

```
        while ((message = in.readLine()) != null) {
```

```
            System.out.println("Message reçu [" + threadName + "] : " + message);
```

```
            switch (message.toLowerCase()) {
```

```
                case "hello":
```

```
                    out.println("Bonjour client !");
```

```
                    break;
```

Soumia KHAY
MBDCC
2025/2026

```
case "time":  
    String date = new SimpleDateFormat("yyyy-MM-dd HH:mm:ss").format(new Date());  
    out.println("Date et heure actuelles : " + date);  
    break;  
case "bye":  
    out.println("Connexion fermée");  
    clientSocket.close();  
    return; // fin du traitement pour ce client  
default:  
    out.println("Message reçu : " + message);  
}  
}  
} catch (IOException e) {  
    System.err.println("Erreur avec le client : " + e.getMessage());  
}  
}  
}
```